Huawei Access Node Hardware/Environmental Fault (ASIC memory parity errors, line card crashes, power supply redundancy loss) Troubleshooting and Resolution SOP

Node: WHI-ACC-81 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_2192

Description

The failure of the Huawei Access Node WHI-ACC-81 at the Whitefield East POP is attributed to hardware/environmental faults within the ASIC memory parity errors, line card crashes, or power supply redundancy loss domains. This is primarily due to the complex interdependencies between the various hardware components and environmental factors that contribute to the overall system reliability. In a Huawei Access Node, the ASIC (Application-Specific Integrated Circuit) plays a critical role in processing and forwarding traffic. Memory parity errors within the ASIC can lead to incorrect data transmission, resulting in network instability. Line card crashes can also occur due to overheating, power surges, or software bugs, causing the node to fail. Additionally, loss of power supply redundancy can cripple the node, leading to service outages. Understanding these underlying mechanics is essential for effective troubleshooting and resolution of hardware/environmental faults in the Huawei Access Node.

Pre-requisite Checks

- Check: Verify the power supply redundancy status using the command "display power supply" and confirm that both power supplies are functioning correctly

- Check: Inspect the line card modules and confirm that they are securely seated and properly connected to the backplane

- Check: Ensure the node's environmental conditions are within the recommended operating range (temperature, humidity, etc.)

- Check: Verify the redundant link status using the command "display link" and confirm that the redundant links are functioning correctly

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display power supply, Expected Output: Power supply status: Power supply 1 is on, Power supply 2 is on, Description: This command is used to verify the power supply redundancy status and confirm that both power supplies are functioning correctly.

Command: display linecard, Expected Output: Linecard 1 is online, Linecard 2 is online, Description: This command is used to verify the line card status and confirm that both line cards are functioning correctly.

Command: display environment, Expected Output: Temperature: 25°C, Humidity: 60%, Description: This command is used to verify the environmental conditions and confirm that they are within the recommended operating range.
```

Resolution Steps

Step 1: Step: Replace faulty power supply, Command: replace power supply 1, Description: Replace the faulty power supply with a spare unit, ensure that the replacement unit is properly installed and configured.

Step 2: Step: Replace faulty line card, Command: replace linecard 1, Description: Replace the faulty line card with a spare unit, ensure that the replacement unit is properly installed and configured.

Step 3: Step: Reboot the node, Command: reboot system, Description: Reboot the node to ensure that the configuration changes take effect and the node is functioning correctly.

Post-Check Verification

```
Command: display power supply, Expected Output: Power supply status: Power supply 1 is on, Power supply 2 is on, Description: Verify that both power supplies are functioning correctly after the resolution steps.

Command: display linecard, Expected Output: Linecard 1 is online, Linecard 2 is online, Description: Verify that both line cards are functioning correctly after the resolution steps.
```

Roll-back Procedures

- Step: Restore original configuration, Command: restore config, Description: Restore the original configuration to the node to prevent further disruption to services.

- Step: Revert to previous version of software, Command: revert software, Description: Revert to the previous version of software to ensure that the node is functioning correctly before the resolution steps.